

SAI VAISHNAVI TIRUPATHI

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SUMMARY

Generalist software engineer and **Final-year CS student** with full-stack engineering depth - built and **deployed 3 production platforms** (88 REST endpoints, 19 DB tables) during a **5-month internship**, and independently **provisioned serverless cloud infrastructure** (AWS, Terraform) as Infrastructure-as-Code. Experience **integrating LLM APIs** for **automated content moderation**, and accelerating development workflows with **AI coding assistants (Claude)** for prototyping, debugging, and test generation.

TECHNICAL SKILLS

Languages: JavaScript, SQL, C++, C
Backend: Node.js, Express.js, RESTful API Design, JWT Authentication, CORS, Data Modeling
Frontend: React.js, Tailwind CSS, JavaScript, React Query
Database & ORM: PostgreSQL, MySQL, Prisma ORM, Redis
Cloud & DevOps: Docker, Terraform, AWS (Lambda, API Gateway, DynamoDB, IAM), GitHub Actions (CI/CD), Render, Neon (cloud Postgres)
AI/ML: LLM API Integration (Google Gemini), Prompt Engineering, Content Moderation Pipelines
Testing: Jest, Supertest, Vitest, React Testing Library, Unit & Integration Testing
Tools: Git, ExcelJS, PDFKit

PROJECT: Serverless Notes API [Github](#)
AWS Lambda, API Gateway, DynamoDB, Terraform, GitHub Actions

- Provisioned a **serverless REST API** (Lambda, API Gateway, DynamoDB) entirely **via Terraform**, using **on-demand billing** and **Infrastructure-as-Code** to **avoid manual resource management**.
- Implemented least-privilege IAM at two layers:** a **Lambda execution role** scoped to exact resource ARNs, and a **custom-scoped deploy policy** restricting the CI/CD user to only project-named resources.
- Automated deployment** with a **GitHub Actions pipeline** running terraform plan on pull requests and **apply on merge** to main.
- Validated the **full CRUD flow end-to-end** (create/read/list/delete) against the **live API Gateway** endpoint.

PROFESSIONAL EXPERIENCE

Software Engineer Intern | Gitam University **March 2026 – July 2026**
(under the mentorship of [Dr.G.Lakshmeeswari](#))

- Owned **3 production-grade full-stack platforms** for GITAM University end-to-end - from database schema to deployment - totalling **88 REST API endpoints** and **19 database tables** across the stack.
- Authored technical documentation** (setup guides, API references, architecture overviews) across **all 3 platforms** for onboarding and maintenance.

MODULE 1: Placements Hub [Github](#) | [Live demo](#)
React, Node.js, Express, PostgreSQL, Prisma, Redis, Docker

- Cut analytics response time **99.9% (728ms → sub-millisecond)** by building a **Redis-ready caching layer** with automatic in-memory fallback.
- Built a placement-prep platform with **33 REST endpoints** and a 12-table PostgreSQL schema, including **10 analytics dashboards (6 Recharts charts)** visualizing company-specific interview trends.
- Wrote **113 unit/integration tests** (65 backend with **Jest/Supertest**, 48 frontend with **Vitest/React Testing Library**), uncovering and **fixing a production bug** in request validation middleware that would have **caused API crashes** on invalid input.
- Containerized** the full stack with **Docker Compose** for **zero-config local dev**, and **deployed to Render** with a GitHub Actions **CI/CD pipeline** on every push.

MODULE 2: GITAM Achievements Portal [Github](#) | [Live demo](#)
Node.js, Express, PostgreSQL (Neon), Vanilla JS, Chart.js, Cloudinary, ExcelJS, PDFKit

- Built a role-based portal (student/faculty/admin) with **24 REST endpoints**, replacing a manual, paper-based certificate approval process.
- Shipped a **secure OTP password reset** (bcrypt-hashed codes, 10-min expiry, 5-attempt lockout) and got the frontend to a **98/100 Lighthouse score** with **0ms blocking time**.
- Traced and **fixed 3 separate production failures** solo: an **SMTP outage** silently **breaking password resets**, disappearing file uploads on redeploy, and a **proxy misconfiguration** that had disabled rate limiting entirely.

MODULE 3: GitamTales [Github](#) | [Live demo](#)
Node.js, Express, PostgreSQL (Neon), JavaScript, Tailwind CSS, JWT

- Built an AI-enabled story-sharing workflow (**31 REST endpoints, 5-table schema**) where students publish project write-ups, achievements, and research reflections to a public feed.
- Integrated Google Gemini API** into the submission pipeline for **automated content moderation**, designing prompts to **flag policy-violating content** (abuse, spam, harassment) across **multiple languages** before publish - **reducing manual review load**.
- Enforced **2-tier RBAC** (user/admin) via JWT + live DB flag checks, and secured password reset with **256-bit cryptographic tokens** (1-hour expiry, anti-enumeration).

EDUCATION

Bachelor of Technology in Computer Science Engineering **2023 – 2027**
GITAM University, Visakhapatnam - 4th Year | CGPA: 8.05/10 | Expected Graduation: April 2027
Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Software Engineering, Design and Analysis of Algorithms